NATIONAL BIOINFORMATICS INFRASTRUCTURE SWEDEN

SciLifeLab

Single cell RNA sequencing data analysis
13-16 April + 27 April, 2026

Jennifer Fransson & Susanne Reinsbach

National Bioinformatics Infrastructure Sweden (NBIS)

'To help the Swedish research community build knowledge in analyzing large and complex omics data and to make bioinformatics easily accessible for life science researchers'

Support

Support services ranging from short consultations to long-term embedded bioinformaticians.

Infrastructure

Providing infrastructure in the form of services, computational resources, tools and guidelines to the life science community.

Training

Training events in advanced and applied bioinformatics.

Need help from NBIS?

- Drop-in sessions
  - Online Tuesdays 14:00
  - On site in Lund, Göteborg, Stockholm
- Consultations
- Support:
  - Long-term support (Peer reviewed)
  - Fee-for-service support, hourly fee
  - Partner projects
- Courses
- Advisory program
- Data management

Who are we?

Course leaders

Jennifer Fransson

Susanne Reinsbach

Teachers

Fariba Roshanzamir

Yuan Li

Åsa Björklund

Stefan Ebmeyer

Aditya Singh

Guest lecturers

Henrik Gezelius
NGI Sweden

Jose Liñares Blanco
University of Heidelberg
EMBL-EBI

Practical information - Zoom

- Please keep your cameras on unless you are having severe network issues
  - Both for lectures and exercises.
- Ask questions:
  - Feel free to unmute and interrupt during lectures!
  - Or by raising your hand in Zoom.
  - Or write in the slack channel #lectures.
- This year’s lectures will not be recorded

Practical information - Zoom

For exercises we will have 4-5 people in each breakout room.

We will post a survey on slack where we want to know which toolkit you wish to practise on to create the groups.

Practical information - Slack

- Communication around the course will be handled via slack.
  - Please add picture and brief description
- Channels:
  - #general – general information about the course
  - #lectures - questions about lectures
  - #exercises – questions during exercises
  - #precourse – questions about the installations
  - #roomX - one for each breakout room for exercises.

Practical information

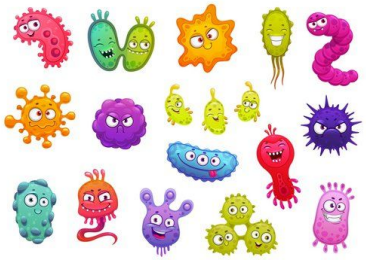
- All materials are available on course website: https://nbisweden.github.io/workshop-scRNAseq/
- Lunches and coffee breaks - please take a minute to step away from the screen!

Course credits

NBIS as an organization cannot give out course credits, only the Universities can.

With your course certificate you should contact the person responsible at your department and usually there is no problem getting credits for the course.

scRNA-seq overview



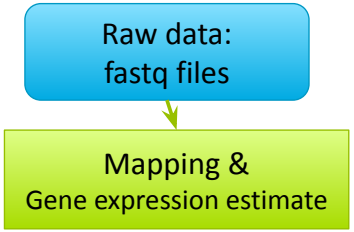
- What method….

- Smartseq2/3
- 10x
- CITE-seq
- Parse
- ….

Raw data: fastq files

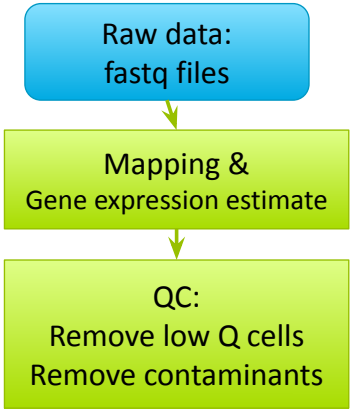
Data analysis

scRNA-seq analysis overview



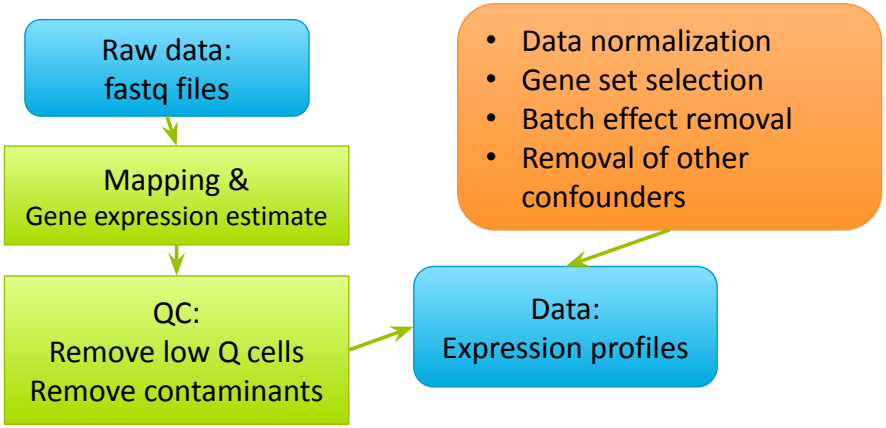
- Cellranger (10x)
- Dropseq - Alevin/Kallisto Bustools (transcriptome only)
- Dropletutils
- nf-core/scrnaseq
- zUMIs
- ….

scRNA-seq analysis overview

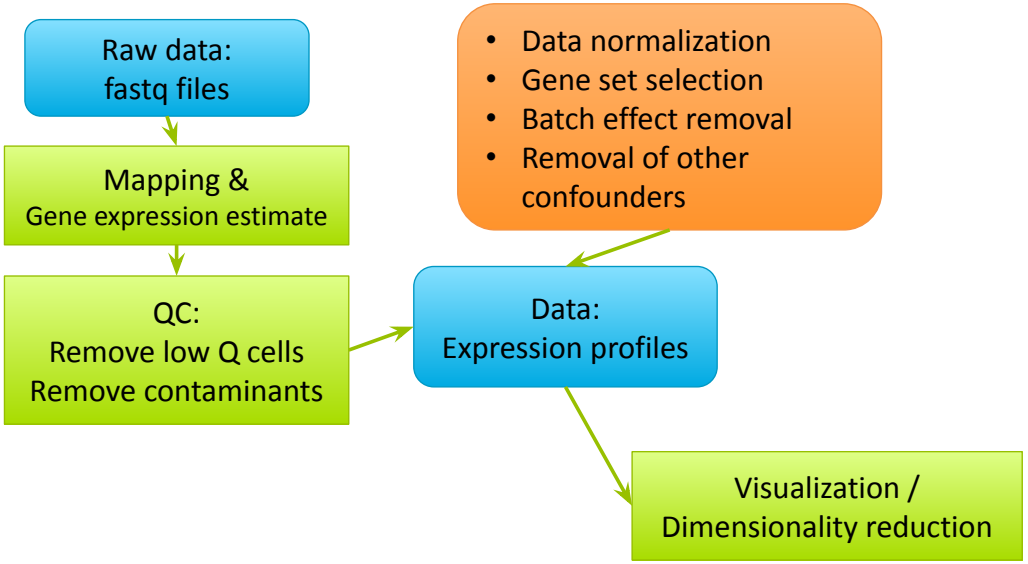


- Low/high number of reads
- Low/high number of genes
- Intronic reads
- Mitochondrial content
- Mapping stats
- Filter genes
- …

scRNA-seq analysis overview

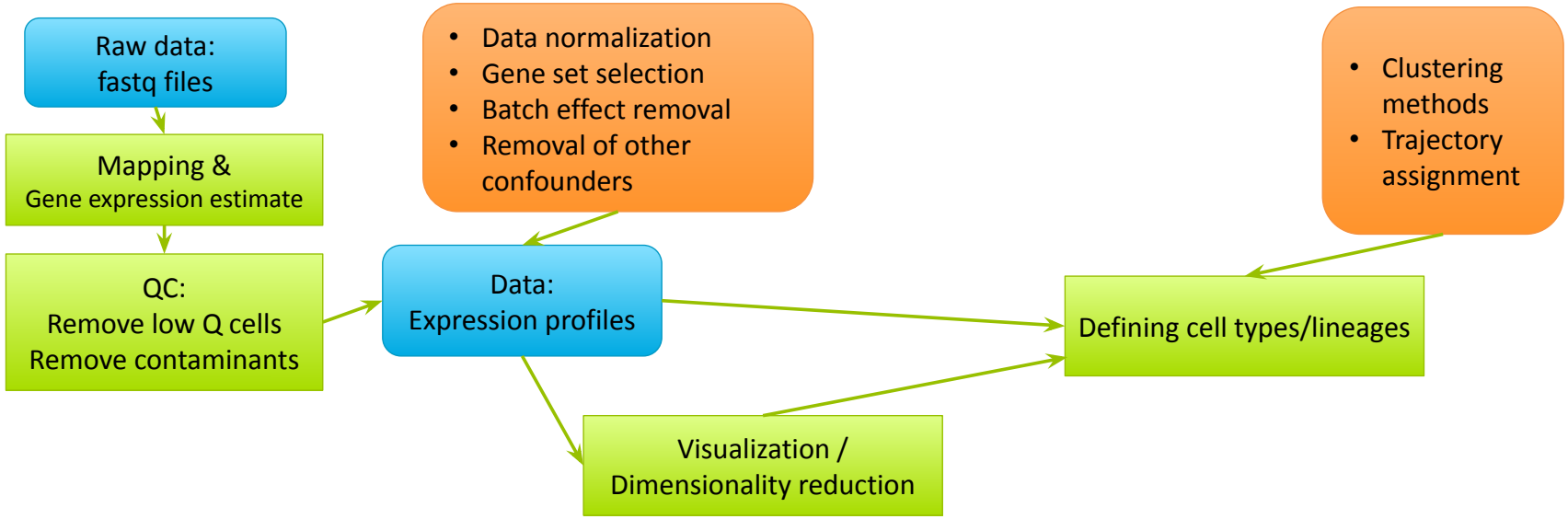


scRNA-seq analysis overview



PCA/ICA/MDS
tSNE
UMAP
Diffusion maps
…

scRNA-seq analysis overview



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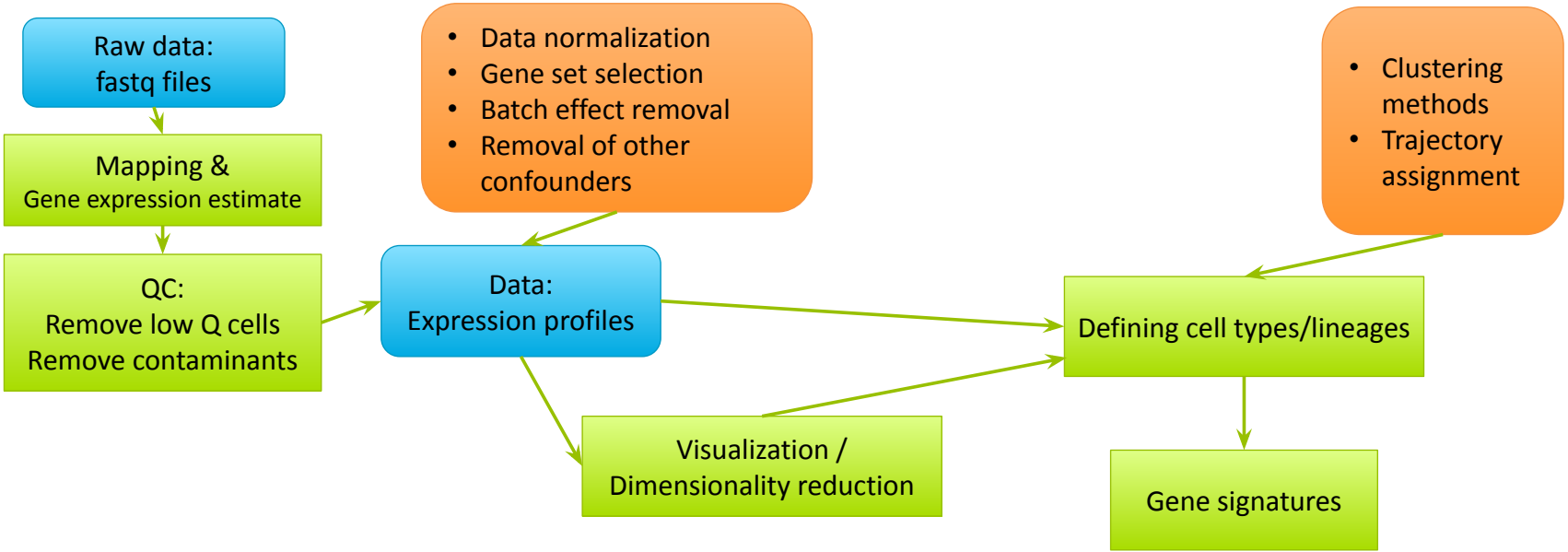

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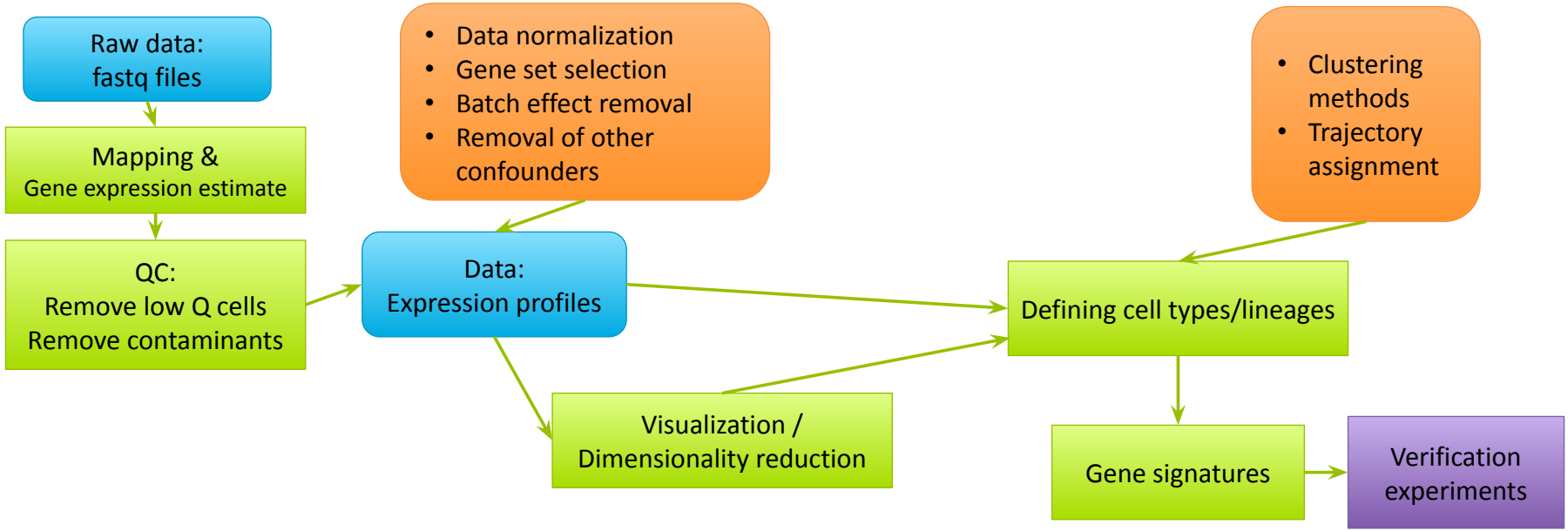
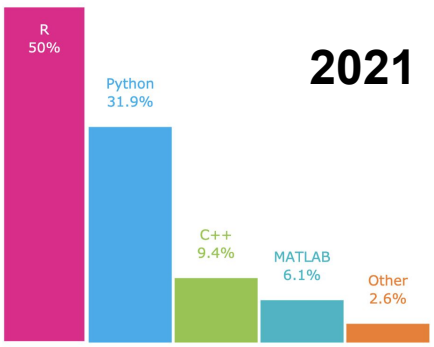
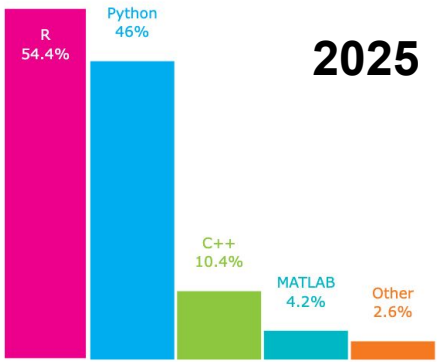
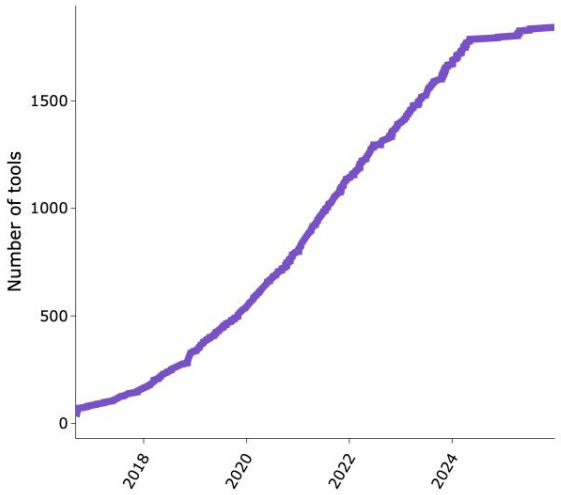


Many available tools for analysis







https://www.scrna-tools.org/

In this course

- We have selected a set of tools/toolkits that we think are relevant for most cases.
- Don’t be afraid of looking for and testing other tools!

Topic	Seurat	Bioconductor	Scanpy
1 Quality Control			
2 Dimensionality reduction			
3 Data integration			
4 Clustering			
5 Differential expression			
6 Celltype prediction			
7 Trajectory inference			

Bring your own data - Monday April 27th

You can work on any dataset of your choice - your own or a public dataset that is relevant to your research (e.g. same tissue, same disease…).

Try out some of the methods you have worked with in the exercises and we will be there to guide you through.

More detailed information will come on April 16th.

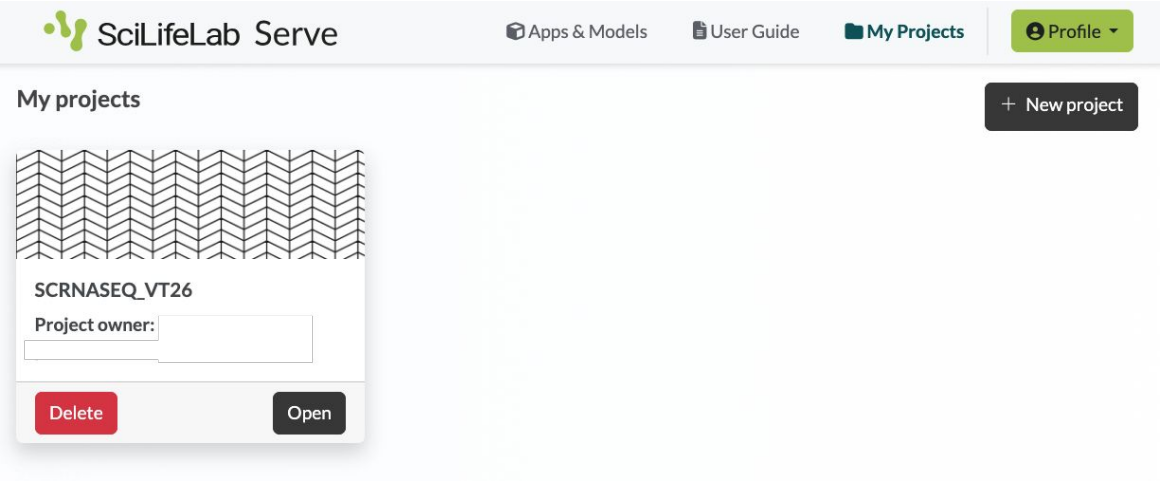
Schedule

https://nbisweden.github.io/workshop-scRNAseq/home_schedule.html

Includes links to PDFs with slides, but some are from last year and will be updated as we give the lectures.

Scilifelab serve account

On serve.scilifelab.se under “My projects”, you should see a project called “SCRNASEQ_VT26”. If this is not the case, please write to us on Slack ASAP.



Scilifelab serve account

Serve is still in beta testing mode, so we want everyone to have Docker/colima installed as a backup solution.

For BYOD the allocated resources may not be enough - more info on Thursday

PLEASE ASK QUESTIONS!

slack